Science

2024/2025 – Middle Grades – Practice 1

Student Instructions:

- Write your complete ID code on your answer sheet.
- If you change your answer, erase well.

1. The gaseous phase is one of the major forms of matter. Which of the following is NOT a characteristic true of gases?
   A. Gases have no definitive shape—they take the shape of their containers
   B. A gas's volume is easily influenced by changes in pressure
   C. The atoms of a gas move slower than the atoms of a liquid at the same temperature
   D. A gas is a type of fluid

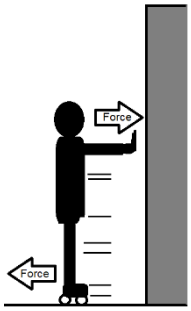
2. An atom of carbon has 6 protons in its nucleus. What quantity is determined by the number of protons in an atom of an element?
   A. atomic weight
   B. average atomic mass
   C. atomic number
   D. atomic mass number

3. Valence electrons are those electrons in the outermost level of an atom. What type of bond is formed when two or more atoms share their valence electrons?
   A. van der Waals bond
   B. covalent bond
   C. metallic bond
   D. ionic bond

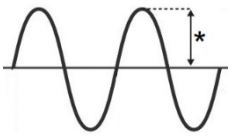
4. Consider the chemical reaction $KBr + AgNO_3 \rightarrow KNO_3 + AgBr$. What type of reaction is this?
   A. single displacement reaction
   B. synthesis reaction
   C. decomposition reaction
   D. double displacement reaction

5. A toy car travels at 2 m/s in a straight line. How far in meters will the toy travel in 5 seconds?
   A. 2.5 m
   B. 10 m
   C. 7 m
   D. 0.4 m

6. The diagram to the right shows a person on roller skates who pushes themselves from a solid wall with a force. Which law of motion explains why the person moves backwards due to the action of an opposite force?
   A. Newton's zeroth law of motion
   B. Newton's first law of motion
   C. Newton's second law of motion
   D. Newton's third law of motion



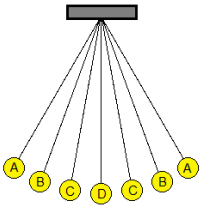
7. Consider the transverse wave given to the right. What is the height of the wave that is marked with an asterisk (*)?



A. amplitude
B. frequency
C. wavelength
D. period

8. A 52-kilogram person stands on a scale to weigh themselves. If the acceleration due to gravity is 10 meters per second squared then how much will the person weigh in newtons?
A. 5.2 N
B. 19.2 N
C. 104 N
D. 520 N

9. A simple pendulum swings freely from a string. Which position in the diagram represents the point in which the pendulum has the greatest kinetic energy?



A. position A
B. position B
C. position C
D. position D

10. Match the method of heat transfer with its example:

x. conduction	1. A metal spoon left in a pan of boiling water gets hot
y. radiation	2. Filling a hot air balloon with hot air causes the balloon to rise
z. convection	3. A pan of water left in direct sunlight becomes warm

A. x-1, y-2, z-3
B. x-1, y-3, z-2
C. x-2, y-1, z-3
D. x-3, y-2, z-1

11. You measure the temperature of a chemical reaction to be 25°C. What is the equivalent temperature in Fahrenheit?
A. 45°F
B. 77°F
C. 147°F
D. 170.6°F

12. The second law of thermodynamics suggests that this quantity is constantly increasing in the Universe. What is the degree of disorder or randomness of a system?
A. temperature
B. enthalpy
C. entropy
D. exergy

13. Match the bacterial shape with its name:

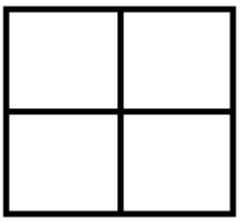
x. coccus	1. corkscrew-shaped
y. spirillum	2. rod-shaped
z. bacillus	3. spherical

A. x-1, y-2, z-3
B. x-2, y-3, z-1
C. x-3, y-1, z-2
D. x-3, y-2, z-1

14. This cellular organelle reads the genetic information from messenger RNA. Which organelles are responsible for the production of proteins?
A. ribosomes
B. nucleoli
C. mitochondria
D. endoplasmic reticula

15. This tissue only transports in one direction. What plant tissue is responsible for the transport of water up from the roots?
A. meristem
B. xylem
C. capillaries
D. phloem

16. In pea plants, purple flowers are dominant to white flowers. What percent of offspring in a cross involving two heterozygous purple-flowered pea plants would also have purple colored flowers? *A blank Punnett square is provided if you want to use it.



A. 25%
B. 50%
C. 75%
D. 100%

17. A template strand of DNA has a short sequence of [*TTGAC*]. What RNA sequence would result from the transcription of this DNA sequence?
A. [*AACTG*]
B. [*GTCAA*]
C. [*TTGAC*]
D. [*AACUG*]

18. A bacterium doubles its genetic material and then slowly divides itself in two to form two identical, yet independent organisms. Which term best describes this process?
A. binary fission
B. mitosis
C. meiosis
D. conjugation

19. Which of the organisms listed can serve as producers in an ecosystem?
A. algae
B. herbivores
C. carnivores
D. scavengers

20. Habitat destruction can lower this quantity. What is the maximum number of organisms that an ecosystem can support?
    A. tidal load
    B. carrying capacity
    C. homeostasis
    D. biotic potential

21. This biome covers approximately 17% of the Earth's surface. Which of the following is the largest terrestrial biome on Earth?
    A. tundra
    B. taiga
    C. temperate deciduous forest
    D. tropical rainforest

22. Sedimentary rock forms when particles settle in an area and compact and harden over time. Which of the following is NOT a sedimentary rock?
    A. sandstone
    B. shale
    C. basalt
    D. limestone

23. The Earth's inner core is solid and its outer core is liquid. Both of these cores are composed primarily of what two metals?
    A. magnesium & aluminum
    B. iron & silicon
    C. aluminum & silicon
    D. iron & nickel

24. According to the National Oceanic and Atmospheric Administration, this layer is on average about 7 miles thick. Which layer of the Earth's atmosphere is responsible for the formation of weather?
    A. stratosphere
    B. thermosphere
    C. exosphere
    D. troposphere

25. These seismic waves are a type of longitudinal wave. Which seismic waves are the first to arrive during an earthquake?
    A. s-waves
    B. r-waves
    C. p-waves
    D. x-waves

26. The mass of a black hole compresses into this region. What lies at the center of a black hole?
    A. singularity
    B. accretion disk
    C. supercore
    D. event horizon

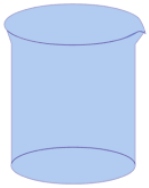
27. This astronaut first entered space in 1983 aboard the space shuttle Challenger. Who was the first American woman in space?
A. Ellen Ochoa
B. Christa McAuliffe
C. Mae Carol Jemison
D. Sally Ride

28. Match the moon with its planet:

x. Ganymede	1. Neptune
y. Titan	2. Saturn
z. Triton	3. Jupiter

A. x-1, y-3, z-2
B. x-3, y-2, z-1
C. x-2, y-1, z-3
D. x-2, y-3, z-1

29. This piece of lab equipment is used primarily to hold liquids. What laboratory apparatus is shown in the diagram?



A. beaker
B. flask
C. graduated cylinder
D. crucible

30. A student sets up an experiment based on the hypothesis "If students receive at least 8 hours of sleep each night the week before a test, then their test scores will be higher than when they receive less than 8 hours of sleep." What serves as the dependent variable in this hypothesis and experiment?
A. quality of sleep
B. amount of sleep
C. tests scores
D. age of the student

31. The "hands only" variety of this procedure will not involve mouth-to-mouth rescue breathing. What safety procedure involves performing rapid and constant chest compressions to maintain a person's blood flow until medical help arrives?
A. SOS
B. EEG
C. CPR
D. EKG

32. Observe the safety symbol provided. What hazard does this symbol warn against?

A. poisonous substance
B. high voltage
C. radiation
D. biohazard

33. This structure develops from the ovule and contains nutrients to nourish the developing young. Which structure contains the developing plant embryo?
A. seed
B. fruit
C. corolla
D. pollen

34. This substance is formed during the light-dependent reactions of photosynthesis. What substance is a waste byproduct that is released by photosynthesis?
A. water
B. nitrogen
C. oxygen
D. carbon dioxide

35. A plant has 6 petals per flower, leaves that have parallel venation, and its vascular bundles are dispersed randomly throughout the stem. What type of plant is this most likely?
A. a monocot
B. a bryophyte
C. a dicot
D. a conifer

36. These animals have been determined to reach a length close to 80 feet long and weigh as much as 300,000 pounds. What is the largest known animal to exist on Earth?
A. brachiosaurus
B. blue whale
C. sperm whale
D. megalodon shark

37. These insects have wings with small colorful scales. What insect order contains the butterflies and moths?
A. Hymenoptera
B. Diptera
C. Lepidoptera
D. Orthoptera

38. These organisms have toe pads instead of claws and their legs reach out from the sides of their bodies rather than from underneath. What type of organism is shown in the diagram?

A. glass lizard
B. fence lizard
C. five-lined skink
D. seal salamander

39. This body form is often described as "umbrella-shaped." What body form is shown by the jellyfish photo?

A. polyp
B. complex
C. sessile
D. medusa

40. Match the disease with the type of organism that causes it:

x. athlete's foot	1. bacterium
y. rabies	2. fungus
z. bubonic plague	3. virus

A. x-1, y-2, z-3
B. x-2, y-1, z-3
C. x-2, y-3, z-1
D. x-3, y-1, z-2

41. The shape of this structure can help classify viruses. What is the outer protein covering of a virus?
   A. capsule
   B. envelope
   C. mycelium
   D. capsid

42. These filaments are adapted to easily absorb and transport water and nutrients throughout a fungus. What are these long filaments that make up a fungal body?
   A. cilia
   B. flagella
   C. mycelia
   D. hyphae

43. This manuscript was funded and published by British astronomer Edmund Halley. Which work by Sir Isaac Newton presented his law of universal gravitation and the laws of motion?
   A. *The Principia*
   B. *On the Revolution of the Spheres*
   C. *Discourse on the Method*
   D. *Novum Organum*

44. This phenomenon was first observed by Heinrich Hertz in 1887, but was not understood until the Twentieth Century. Albert Einstein won the 1921 Nobel Prize in Physics for his explanation of what phenomenon?
   A. photoelectric effect
   B. Theory of special relativity
   C. mass-energy equivalence ($E = mc^2$)
   D. Theory of general relativity

45. This prototype weapon contained plutonium rather than uranium. What was the nickname given to the first working nuclear bomb that was detonated at the Trinity site at Alamogordo, New Mexico?
   A. Thin Man
   B. the Gadget
   C. Little Boy
   D. Fat Man

46. Match the invention with its inventor:

x. nylon	1. Wallace Carothers
y. world wide web	2. Douglas Engelbart
z. computer mouse	3. Tim Berners-Lee

   A. x-1, y-3, z-2
   B. x-2, y-3, z-1
   C. x-3, y-1, z-2
   D. x-3, y-2, z-1

47. A deficiency in this vitamin may result in night blindness. What vitamin is needed for proper embryonic development, maintaining a healthy immune system, and the maintenance of healthy vision?
   A. Vitamin E
   B. Vitamin A
   C. Vitamin D
   D. Vitamin B_6

48. This acid is produced by the parietal cells of the stomach. Which acid is the primary component of gastric acid?
    A. hydrochloric acid
    B. sulfuric acid
    C. nitric acid
    D. phosphoric acid

49. This informative diagram was released in 2011 to focus on improving Americans' diets. What diagram suggests that you should "make half your plate fruits and vegetables"?
    A. Trophic pyramid
    B. MyPlate
    C. MyPyramid
    D. Food Guide Pyramid

50. This condition occurs due to a lengthening of the eye such that images are not in focus when projected onto the retina. What condition is better known as near-sightedness?
    A. glaucoma
    B. myopia
    C. astigmatism
    D. hyperopia

Science – Middle Grades – Practice 1
2024-25 Governor's Cup

1. C
2. C
3. B
4. D
5. B
6. D
7. A
8. D
9. D
10. B
11. B
12. C
13. C
14. A
15. B
16. C
17. D
18. A
19. A
20. B
21. B
22. C
23. D
24. D
25. C
26. A
27. D
28. B
29. A
30. C
31. C
32. D
33. A
34. C
35. A
36. B
37. C
38. D
39. D
40. C
41. D
42. D
43. A
44. A
45. B
46. A
47. B
48. A
49. B
50. B